

1 Hermitian operator

Hermitian operator: $A = A^*$.

Spectral decomposition of a Hermitian operator: $A = \sum_{\lambda, |\lambda\rangle} \lambda |\lambda\rangle \langle \lambda|$. It satisfies that $\sum |\lambda\rangle \langle \lambda| = I$ and $\langle \lambda_i | \lambda_j \rangle = \delta_{ij}$.

Schrödinger equation: $i\hbar \frac{\partial |\psi\rangle}{\partial t} = H |\psi\rangle$.

When $H(t) = H$, we have $|\psi(t)\rangle = \exp(-iH(t - t_0)/\hbar) |\psi(t_0)\rangle$.

We call $U(t, t_0)$ that $|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle$ as evolution operator here. It's unitary and norm is reserving during evolution.

projector: $P = |\lambda\rangle \langle \lambda|$. $P \geq 0, P^2 = P, \sum P_i = I$.

2 Finite-dim quantum system and Continuous-variable system

Finite-dim quantum system: dim is finite.

e.g. when $d = 2$, maybe: 1) spin-1/2 particles. $\uparrow, \downarrow$. 2) two level atoms. 3) photon polarization. 4) quantum dots (nano structures). 5) Josephson junction (superconducting device). 6) Neutrino oscillations.

2-D system: $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, |\psi\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \exp(ir)(\cos \frac{\theta}{2} |0\rangle + \exp(i\phi) \sin \frac{\theta}{2} |1\rangle)$

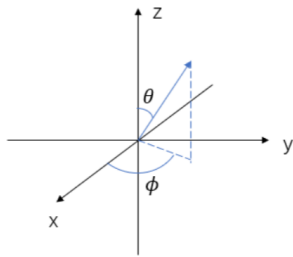
Basis states $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} \quad |c_0|^2 + |c_1|^2 = 1$$

$$= e^{ir} (\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle)$$

r: Globe phase – not important ϕ : relative phase θ : angle $\Rightarrow 3$ real numbers

Geometric representation: $\Omega = (\theta, \phi)$



Spin operator $\vec{S} \equiv \frac{\hbar}{2} \vec{\sigma}$ (definition)

Basis vector of $\sigma_x, \sigma_y, \sigma_z$ form $\sigma_i |\psi\rangle = \lambda_i |\psi\rangle$

one finds $\lambda_i = \pm 1$ (as $\sigma_i^2 = I$)

eigenvectors:

$$\begin{array}{ccc} \sigma_z \begin{array}{l} |0\rangle \\ |1\rangle \end{array} & \sigma_x \begin{array}{l} (|0\rangle + |1\rangle)/\sqrt{2} \\ (|0\rangle - |1\rangle)/\sqrt{2} \end{array} & \sigma_y \begin{array}{l} (|0\rangle + i|1\rangle)/\sqrt{2} \\ (|0\rangle - i|1\rangle)/\sqrt{2} \end{array} \end{array}$$

Pauli matrix: $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \text{diag}\{1, -1\}$.

$\sigma^* = \sigma, \sigma^2 = I, \sigma_x \sigma_y = -\sigma_y \sigma_x = i\sigma_z$.

I and Pauli matrices construct a basis for 2-D Hermitian operators.

Continuous-variable system: infinite-dim and some operator has continuous eigenspectrum (eigenvalues).

e.g. $\hat{x} = \int x |x\rangle \langle x| dx, \hat{p} = \int p |p\rangle \langle p| dp, \langle x|x'\rangle = \delta(x - x'), \int |x\rangle \langle x| dx = \hat{I}$.

we can prove that $|p\rangle = \int \frac{\exp(ipx/\hbar)}{\sqrt{2\pi\hbar}} |x\rangle dx$.

Under the basis of $\hat{x}$, we have $\hat{p} \Leftrightarrow -i\hbar \frac{d}{dx}$. Similarly $\hat{x} \Leftrightarrow i\hbar \frac{d}{dp}$.

3 Uncertainty relation, SP and HP

Uncertainty relation: If $[A, B] \neq 0$, we have $\langle (\Delta A)^2 \rangle \langle (\Delta B)^2 \rangle \geq \frac{1}{4} |\langle [A, B] \rangle|^2$. Here $\Delta A = A - \langle A \rangle$.

Time evolution in Schrödinger picture: $i\hbar \frac{\partial |\psi\rangle}{\partial t} = H |\psi\rangle$.

e.g. In the CV system, we have $H = \hat{p}^2/2m + V(\hat{x})$.

Time evolution in Heisenberg picture: $i\hbar \frac{\partial A_H}{\partial t} = [A_H, H]$. constant state but time-related operator.

Case a: in the coordinate representation.

$$i\hbar \frac{\partial \langle x|\psi\rangle}{\partial t} = \langle x|\hat{H}(\hat{x}, \hat{p})|\psi\rangle$$

$$\langle x|\psi\rangle = \psi(x)$$

$$\langle x|\hat{H}(\hat{x}, \hat{p})|\psi\rangle = H\left(x, -i\hbar \frac{\partial}{\partial x}\right)\psi(x) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x)$$

$$\text{So } \Rightarrow i\hbar \frac{\partial \psi(x)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x) \quad (\text{wave equation})$$

(Schrödinger equation in the coordinate representation.)

$$\stackrel{3D}{\Rightarrow} i\hbar \frac{\partial \psi(\vec{r})}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla_{\vec{r}}^2 + V(\vec{r})\right]\psi(\vec{r})$$

Case b: in the momentum $|p\rangle$ representation

$$i\hbar \frac{\partial \langle p|\psi\rangle}{\partial t} = \langle p|H(\hat{x}, \hat{p})|\psi\rangle$$

$$\langle p|\psi\rangle = \phi(p)$$

$$\langle p|H(\hat{x}, \hat{p})|\psi\rangle = H\left(i\hbar \frac{\partial}{\partial p}, p\right)\phi(p)$$

$$\text{So, } i\hbar \frac{\partial \phi(p)}{\partial t} = \left[-\frac{p^2}{2m} + V\left(i\hbar \frac{\partial}{\partial p}\right)\right]\phi(p) \quad (\text{momentum-basis Schrödinger Eq.})$$

3. Evolution in the Heisenberg picture

	State	Operator	Phys really observed property
SP	$ \psi(t)\rangle_s = \hat{U}(t) \psi(0)\rangle$	$A_S = A$ (t-independent)	$\langle A \rangle = {}_s\langle \psi(t) A \psi(t)\rangle_s$ $\Updownarrow$ Independent of picture (frame) $\langle A \rangle = {}_H\langle \psi A(t) \psi\rangle_H$
HP	$ \psi\rangle_H \equiv \psi(0)\rangle$	$A_H(t) = \hat{U}^\dagger(t)A\hat{U}(t)$	

$$\text{From } i\hbar \frac{\partial \langle \psi(t)|}{\partial t} = i\hbar \frac{\partial \langle \psi(0)|}{\partial t} |\psi(0)\rangle = H|\psi(t)\rangle = H\hat{U}(t)|\psi(0)\rangle$$

$$\text{Valid for any } |\psi(0)\rangle \Rightarrow \text{So } i\hbar \frac{\partial \hat{U}}{\partial t} = H\hat{U}$$

$$\text{As } A_H(t) = \hat{U}^\dagger(t)A\hat{U}(t)$$

$$i\hbar \frac{dA_H}{dt} = i\hbar \left[\frac{dU^\dagger}{dt}AU + U^\dagger A \frac{dU}{dt} \right] = [-U^\dagger HAU + U^\dagger AHU]$$

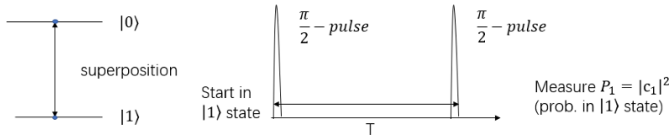
(Assumed $\frac{\partial A}{\partial t} = 0$) $[U, H] = 0$

$$\downarrow$$

$$= [-HU^\dagger AU + U^\dagger AUH]$$

$$= A_H(t)H - HA_H(t) = [A_H, U]$$

$$\text{So } \begin{cases} i\hbar \frac{dA_H}{dt} = [A_H, H] & (\text{Heisenberg equation}) \\ |\psi\rangle_H = |\psi(0)\rangle \end{cases}$$



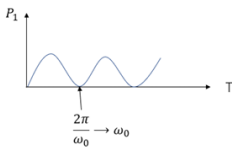
$$|1\rangle \xrightarrow{\frac{\pi}{2}\text{-pulse } (\sigma_y\text{-basis})} \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle)$$

$$\xrightarrow{\text{free evolution for a time } T (\sigma_z\text{-basis})} \frac{1}{\sqrt{2}}(|1\rangle e^{\frac{i\omega_0 T}{2}} + |0\rangle e^{-\frac{i\omega_0 T}{2}})$$

$$\xrightarrow{\frac{\pi}{2}\text{-pulse again } (\sigma_y\text{-basis})} \frac{1}{2}[(|1\rangle + |0\rangle)e^{\frac{i\omega_0 T}{2}} + (|0\rangle - |1\rangle)e^{-\frac{i\omega_0 T}{2}}]$$

$$= |0\rangle \cos \frac{\omega_0 T}{2} + i|1\rangle \sin \frac{\omega_0 T}{2}$$

$$\xrightarrow{\text{Measure } P_1} P_1 = \sin^2 \frac{\omega_0 T}{2}$$

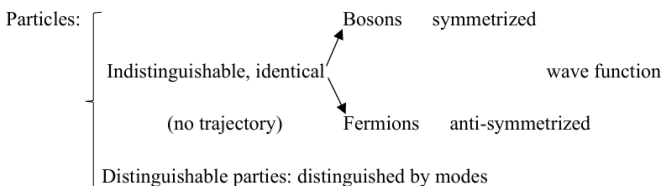


$$T \xrightarrow{P_1} \omega_0 \quad (\text{frequency standard})$$

$$\omega_0 \xrightarrow{P_1} T \quad (\text{clock})$$

§ 2.5 Multi-partite quantum systems, density matrix, and quantum entanglement

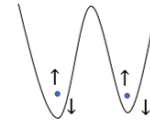
1. Multi-partite quantum systems



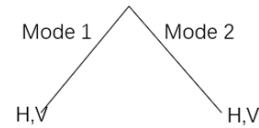
Ex.

1) Spin $\uparrow, \downarrow$ particle

2) Lattice spin



3) Photon



Multi-partite systems: The Hilbert-space can be decomposed as

$$H = H_1 \otimes H_2 \otimes \dots \otimes H_n$$

Tensor product

Basis vectors can be written as tensor products

Ex.

Basis vectors

1) 1 qubit (spin) $|0\rangle, |1\rangle$ dim-2

2) 2 qubits $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ dim-2²

3) n qubits $|00 \dots 1\rangle$ dim-2ⁿ

n bits exponentially expanded Hilbert space dimensions (intractable)

Remarks: 1) Many-particle systems, when particles are distinguished by modes, can be considered as multi-partite systems

2) Particular multi-partite state

4 Quantum dynamics of 2D system

2-D Hamiltonian: $H = \hbar(\alpha I + \frac{\omega}{2} \vec{n} \cdot \vec{\sigma})$. $|\vec{n}| = 1$, α, ω independent of t .

Its evolution: $U(t) = \exp(-iHt/\hbar) = \exp(-i\alpha t)(\hat{I} \cos(\omega t/2) - i\vec{n} \cdot \vec{\sigma} \sin(\omega t/2))$.

Atomic clocks: $\Delta E = E_{|0\rangle} - E_{|1\rangle} = \hbar\omega$. Take $E_{|0\rangle} + E_{|1\rangle} = 0$, we have $H = \text{diag}\{E_{|0\rangle}, E_{|1\rangle}\} = \frac{\hbar\omega}{2} \sigma_z$.

Then $U = \exp(-\frac{i\omega t \sigma_z}{2}) = \exp(-\frac{i\omega t}{2})E_{00} + \exp(\frac{i\omega t}{2})E_{11}$, which gives a clock.

Ramsey method: Method to calculate ω_0 . Suppose $|1\rangle$ in the beginning.

Step 1: give a $\pi/2$ pulse(along σ_y) so that become $\frac{|0\rangle + |1\rangle}{\sqrt{2}}$.

Step 2: free evolution so that become $\frac{|0\rangle \exp(-i\omega_0 t/2) + |1\rangle \exp(i\omega_0 t/2)}{\sqrt{2}}$.

Step 3: give a $\pi/2$ pulse(along σ_y) so that become $|0\rangle \cos(\omega_0 t/2) + i|1\rangle \sin(\omega_0 t/2)$.

Step 4: The possibility to be $|1\rangle$ would be $\sin^2(\omega_0 t/2)$. It'd be 0 firstly when $t = 2\pi/\omega_0$.

5 Multi-partite quantum systems, density matrix

Multi-partite quantum systems: multi-particles.

Product state is given by $\otimes$ of substates. Its degree of freedom is only $2n$.

Non product state for example: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

trace: $\text{tr} A = \sum_j A_{jj}$. Under a basis of $|\psi_i\rangle$, we have $\text{tr} A = \sum_i \langle \psi_i | A | \psi_i \rangle$.

Density matrix operator: $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$.

Prop 1: ρ is density operator iff $\text{tr} \rho = 1 \wedge \rho \geq 0$.

Prop 2: $\text{tr}(\rho^2) = 1$. And $\text{tr}(\rho^2) = 1$ iff ρ is a pure state that $\rho = |\psi\rangle \langle \psi|$.

2) Reduced state interpretation

Particles

1

2

other side of galaxy

$|\psi\rangle_{12} = c_0|00\rangle + c_1|11\rangle$

What is the state of particle 1?

For any operator A_1 of particle 1,

$\langle A_1 \rangle = {}_{12} \langle \psi | A_1 | \psi \rangle_{12} = \text{tr}(A |\psi\rangle_{12} {}_{12} \langle \psi |)$

$\text{tr} = \text{tr}_{12} = \text{tr}_1 \text{tr}_2$ (partial trace)

Def. of “partial trace”, For operator B_{12}

$\text{tr}_2(B_{12}) \equiv \sum_{\mu_{2i}}$

$\langle \mu_{2i} | B_{12} | \mu_{2i} \rangle$

an operator on particle 1

basis vector for particle 2

so $\langle A_1 \rangle = \text{tr}_1[A_1 \text{tr}_2(|\psi\rangle_{12} {}_{12} \langle \psi |)] = \text{tr}(\rho_1 A_1)$, where $\rho_1 \equiv \text{tr}_2(|\psi\rangle_{12} {}_{12} \langle \psi |)$

Reduced state ρ_1 : completely characterize the state of the particle 1.

In general, ρ_1 is a density operator

mixed state superoperator

ensemble Rep. $\rho_A = \sum_i |i\rangle \langle i| \Leftrightarrow$ Kraus Rep. $\hat{\mathbb{S}}(\rho_A) = \sum_i M_\mu \rho M_\mu^\dagger$

purification $\rho_A = \text{tr}_B(|\psi\rangle_{AB} {}_{AB} \langle \psi|) \Leftrightarrow \hat{\mathbb{S}}(\rho_A) = \text{tr}_B(U_{AB}(\rho_A \otimes |E\rangle_B {}_B \langle E|)U_{AB}^\dagger)$

3. Derivation of the Master Equation from Kraus rep.

$\rho(t + \delta t) = \hat{\mathbb{S}}(\rho(t)) = \sum_\mu M_\mu \rho(t) M_\mu^\dagger$ (No initial entanglement, no memory for B)

Assume δt is small, $\rho(t + \delta t) = \rho(t) + \delta t \dot{\rho}$

One M_μ must be close to I, let $M_0 = I + \delta t(-iH + K)$

$\sum_\mu M_\mu^\dagger M_\mu = \frac{M_0^\dagger M_0}{I + 2\delta t \cdot K} + \sum_{\mu \neq 0} M_\mu^\dagger M_\mu = I$

So for $\mu \neq 0$ $M_\mu \sim \sqrt{\delta t}$ Let $M_\mu = L_\mu \sqrt{\delta t}$

$(\sum_\mu L_\mu^\dagger L_\mu + 2K)\delta t = 0$, $K = -\frac{1}{2} \sum_\mu L_\mu^\dagger L_\mu$

Let $\delta t \rightarrow 0$,

Hamiltonian

Master Eq.

jump operators

Lindblad form

$\dot{\rho} = \frac{\rho(t + \delta t) - \rho(t)}{\delta t} = -i[H, \rho] + \sum_\mu [L_\mu \rho L_\mu^\dagger - \frac{1}{2} L_\mu^\dagger L_\mu \rho - \frac{1}{2} \rho L_\mu^\dagger L_\mu]$

6 Superoperator

Superoperator: a map from operator to operator.

Evolution superoperator: $\hat{\$} : \rho \rightarrow \rho' \Leftrightarrow \rho' = \hat{\$}(\rho)$.

It's linear. It preserves hermiticity. $\text{tr } \rho = \text{tr } \rho' = 1$ $\rho' \geq 0$ if $\rho \geq 0$. It's completely positive (strong) (any extensive $\hat{\$} \otimes \hat{I}_B$ is positive).

Transposition superoperator: $\hat{T} : \rho \rightarrow \rho^T$.

It's a positive but not-completely-positive superoperator.

Kraus theorem: For any superoperator satisfying:

- 1). linear. 2). preserve hermiticity. 3). preserve trace. 4). completely positive.

we have an operator-sum representation: $\hat{\$}_A(\rho_A) = \sum_{\mu} M_{\mu} \rho_A M_{\mu}^*$ where $\sum_{\mu} M_{\mu}^* M_{\mu} = I_A$.

Proof: completely positive gives Stinespring: $\hat{\$}_A(\rho_A) = \text{tr}_B(V \rho_A V^*)$ where V is in $A \rightarrow A \otimes B$.

Then it becomes $\hat{\$}_A(\rho_A) = \text{tr}_B(U(\rho_A \otimes |E\rangle_B \langle E|)U^*)$. Take $M_{\mu} = \langle \mu |_B U_{AB} |E\rangle_B$ (Only multiply in the B part). $\square$.

Here we have at most $\dim^2(H_A) = \dim(H_A \otimes H_A)$ ones of M_{μ} .

If here's another possible $N_{\nu} = \langle \nu |_B U_{AB} |E\rangle_B = \sum_{\mu} \langle \nu | \mu \rangle M_{\mu}$. So they can be transfer by a unitary matrix.

Inverse theorem of Kraus: Relation between operator-sum and unitary representation.

Th: Any evolution $\hat{\$}$ on density operator ρ_A with $\hat{\$}(\rho_A) = \sum_{\mu} M_{\mu} \rho_A M_{\mu}^+$ can be implemented with a unitary evolution U_{AB} in the extended space $H_A \otimes H_B$ with

$$\text{tr}_B[U_{AB}(\rho_A \otimes |E\rangle_B \langle E|)U_{AB}^+] = \sum_{\mu} M_{\mu} \rho_A M_{\mu}^+$$

any ancilla state

Proof: Assume $\rho_A = \sum_i p_i |\phi_i\rangle \langle \phi_i|$

Define U_{AB} as $U_{AB}(|\phi_i\rangle_A \otimes |E\rangle_B) \equiv \sum_{\mu} M_{\mu} |\phi_i\rangle_A \otimes |\mu\rangle_B$
(orthonormal states in H_B)

Such U_{AB} exists because the above mapping is inner-product preserving

$$U_{AB}(|\psi_j\rangle_A \otimes |E\rangle_B) = \sum_{\nu} M_{\nu} |\psi_j\rangle_A \otimes |\nu\rangle_B$$

Inner product

$$\text{right} = \sum_{\mu} {}_A \langle \psi_j | M_{\mu}^+ M_{\mu} | \phi_i \rangle_A = {}_A \langle \psi_j | \phi_i \rangle_A = \text{left}$$

With such a U_{AB}

5. Examples of superoperator evolution: quantum channels.

$$\rho \rightarrow \hat{\$}(\rho) \qquad \rho \xrightarrow{\hat{\$}} \rho' = \hat{\$}(\rho)$$

1) Depolarizing channel: $1 - p$ no error $\hat{I}$

For a qubit state

$$p \left\{ \begin{array}{l} \frac{p}{3} \\ \frac{p}{3} \\ \frac{p}{3} \end{array} \right. \left\{ \begin{array}{l} \text{a bit flip } \sigma_x = X \\ \text{a phase flip } \sigma_z = Z \\ \text{a bit - phase flip } \sigma_y = Y = iXZ \end{array} \right. \text{error}$$

So Krause Rep. $\rho \rightarrow \hat{\$}(\rho) = \sum_{i=0,1,2,3} M_{\mu} \rho M_{\mu}^+$,

$$\text{with } M_0 = \sqrt{1-p}I, M_1 = \sqrt{\frac{p}{3}}X, M_2 = \sqrt{\frac{p}{3}}Y, M_3 = \sqrt{\frac{p}{3}}Z$$

$$\rho' = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

In Bloch - sphere Rep. $\rho = \frac{1}{2}(I + \vec{p} \cdot \vec{\sigma})$ $\left(\begin{smallmatrix} XXX=X \\ XYY=Y \end{smallmatrix} \right)$

$$\rho' = \frac{1}{2}(I + \vec{p}' \cdot \vec{\sigma}), \text{ with } \vec{p}' = (1 - \frac{4}{3}p)\vec{p}$$

$p = \frac{3}{4}$ complete randomized shrinking ball

$$\dot{\rho} = \gamma(X\rho X + Y\rho Y + Z\rho Z - 3\rho)$$

2) Phase-damping channel

$$\left\{ \begin{array}{l} 1 - p, \text{ no error, } I \\ p, \text{ a phase flip error, } \sigma_z = Z \end{array} \right.$$

Krause operator $M_0 = \sqrt{1-p}I, M_1 = \sqrt{p}Z$

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}$$

$$\hat{\$}(\rho) = (1-p)\rho + pZ\rho Z = \begin{pmatrix} \rho_{00} & (1-2p)\rho_{01} \\ (1-2p)\rho_{10} & \rho_{11} \end{pmatrix}$$

dephasing decoherence

$$\begin{aligned} \text{tr}_B[U_{AB}(\rho_A \otimes |E\rangle_B \langle E|)U_{AB}^+] &= \sum_i p_i \{ \text{tr}_B[\sum_{\mu, \nu} M_{\mu} |\phi_i\rangle_A \otimes |\mu\rangle_B \langle \nu| \langle \phi_i| M_{\nu}^+] \} \\ &= \sum_{\mu} M_{\mu} \left(\sum_i p_i |\phi_i\rangle_A \langle \phi_i| \right) M_{\mu}^+ = \sum_{\mu} M_{\mu} \rho_A M_{\mu}^+ \end{aligned}$$

7 Entanglement theory

Bi-partite pure state Quantum entanglement: If $|\psi\rangle_{AB} \neq |\psi\rangle_A \otimes |\psi\rangle_B$, we call it to be entanglement.

Von Neumann’s entropy: $S(\rho) = -\text{tr}(\rho \log \rho)$.

$S(U\rho U^*) = S(\rho), S(|\psi\rangle\langle\psi|) = 0, S(\lambda_1\rho_1 + \lambda_2\rho_2) \geq \lambda_1 S(\rho_1) + \lambda_2 S(\rho_2).$

$S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B)$ (equal iff $\rho_{AB} = \rho_A \otimes \rho_B$), $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$

$S(\rho_{AB}) \geq |S(\rho_A) - S(\rho_B)|$

We can measure entanglement for pure state $\rho_{AB} = |\psi\rangle_{AB}$ by $E(|\psi\rangle_{AB}) = S(\rho_A) = S(\rho_B)$, as $S(\rho_{AB}) = 0$.

Schmidt decomposition: for a pure state $\rho_{AB} = |\psi\rangle_{AB}$, we have $|\psi\rangle_{AB} = \sum_i \sqrt{p_i} |\alpha_i\rangle_A \otimes |\beta_i\rangle_B$ with $\sum p = 1$.

Here we have $\rho_A = \sum_i p_i |\alpha_i\rangle\langle\alpha_i|$ and $\rho_B = \sum_i p_i |\beta_i\rangle\langle\beta_i|$.

In that way, we have $\rho_A = \text{tr}_B(|\psi\rangle_{AB}\langle\psi|_{AB})$ for any ρ_A , $|\psi\rangle_{AB}$ is called purification of ρ_A .

LOCC: Local Operation and Classical Communications. Can’t increase entanglement.

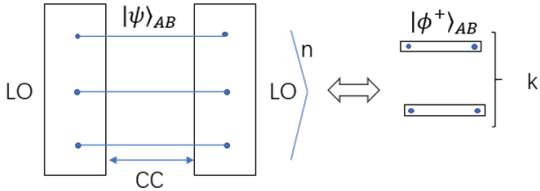
Single-copy operation: $|\psi\rangle_{AB} \xrightarrow{LOCC} \chi |\phi^+\rangle_{AB}$. χ quantifies entanglement.

Asymptotic operation: $|\psi\rangle_{AB}^{\otimes n} \xrightarrow{LOCC} |\phi^+\rangle_{AB}^{\otimes k}$. $\chi = k/n$ here.

How to quantify entanglement? Compare it to $|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Single-copy operation: $|\psi\rangle_{AB} \xrightarrow{LOCC} \chi |\phi^+\rangle_{AB}$, χ : quantifies entanglement

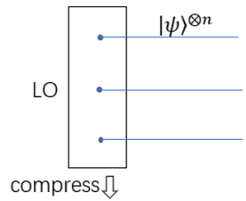
Asymptotic operation: $|\psi\rangle_{AB}^{\otimes n} \xrightarrow{LOCC} |\phi^+\rangle_{AB}^{\otimes k}$ $\chi = \frac{k}{n}$

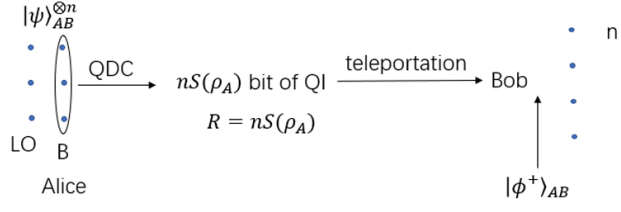


The ratio $E_F(|\psi\rangle_{AB}) \equiv \frac{k_{min}}{n}$,

$E_F(|\psi\rangle_{AB}) = E_D(|\psi\rangle_{AB}) = S(\rho_A)$

Proof idea: { Quantum data compression
+ Quantum teleportation





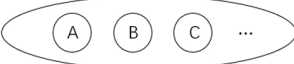
Entanglement of formation: k_{min} is the least number of $|\phi^+\rangle$ to make n copies of $|\psi\rangle_{AB}$ through LOCC. $E_F = k_{min}/n$.

Entanglement of distillation: k_{min} is the maximum number of $|\phi^+\rangle$ that can be distilled from n copies of $|\psi\rangle_{AB}$ through LOCC. $E_D = k_{max}/n$. $E_D \leq E_F$ as LOCC can’t increase entanglement.

Entanglement of bi-partite pure states: $E_D(|\psi\rangle_{AB}) = E_F(|\psi\rangle_{AB}) = S(\rho_A)$.

Multi-partite entanglement: No simple answer.

4. Multi-partite entanglement



How to characterize?

In general, no simple answer.

Remarks: 1) Different types of entanglement order, no unique measure.

2) Transfer to bi – partite problem

* Block entanglement * Reduced density ρ_{AB} * Localizable entanglement

Proof: (Partial) Peres If ρ_{AB} is separable $\Rightarrow \rho_{AB}^T > 0$

$$\begin{aligned} \rho_{AB} &= \sum_i p_i \rho_{iA} \otimes \rho_{iB} \\ \rho_{iA} &= \sum_j q_{ij} |e_{ij}\rangle_A \langle e_{ij}| \\ \rho_{iB} &= \sum_k q_{ik} |\mu_{ik}\rangle_B \langle \mu_{ik}| \\ &= \sum_{i,j,k} p_i q_{ij} q_{ik} |e_{ij}\rangle_A \langle e_{ij}| \otimes |\mu_{ik}\rangle_B \langle \mu_{ik}| \\ &\quad \downarrow \quad \downarrow \\ &\quad i' \quad j' \end{aligned}$$

$$\xrightarrow{\text{simplify notation without loss of generality}} \rho_{AB} = \sum_i p_i |e_i\rangle_A \langle e_i| \otimes |\mu_i\rangle_B \langle \mu_i|$$

Bi-partite mixed state Quantum entanglement: If $\rho_{AB} \neq \sum_i p_i \rho_{iA} \otimes \rho_{iB}$ in any way, we call it entangled state. If not, we call it a separable state.

Partial transpose (Peres – Horodecki) criteria: For any separable state ρ_{AB} , the partial transpose $\rho_{AB}^{T_B}$ is positive. This is called Positive Partial Transpose State (PPT State).

It’s a necessary and sufficient condition for 2×3 and 2×2 cases, but not for 3×3 (bound entanglement).

$$\begin{aligned} \rho_{AB}^{T_B} &= \sum_i p_i \rho_{iA} \otimes \rho_{iB}^{T_B} = \sum_i p_i \rho_{iA} \otimes \rho_{iB}^* \\ \rho_{iA} &= \sum_j q_{ij} |e_{ij}\rangle_A \langle e_{ij}| \\ \rho_{iB}^* &= \sum_k q_{ik} |\mu_{ik}^*\rangle_B \langle \mu_{ik}^*| \\ &= \sum_{ijk} p_i q_{ij} q_{ik} |e_{ij}\rangle_A \langle e_{ij}| \otimes |\mu_{ik}^*\rangle_B \langle \mu_{ik}^*| > 0 \\ &\quad \downarrow \\ &\text{ensemble decomposition} \end{aligned}$$

Horodecki $\rho_{AB}^{T_B} > 0 \Rightarrow \rho_{AB}$ is separable for $2 \times 2, 2 \times 3$ systems.

Ex. 1) $|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$$\begin{aligned} \rho_{AB}^{T_B} &= \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11| + |01\rangle\langle 10| + |10\rangle\langle 01|) \\ &\rightarrow \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix} \\ \det(\rho_{AB}^{T_B}) &= -1 \text{ entangled} \end{aligned}$$

2) $\rho_{AB} = p|\phi^+\rangle_{AB}\langle\phi^+| + \frac{(1-p)}{4}I_{AB}$ (Werner state)

$$\rho_{AB}^{T_B} = \frac{p}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \frac{(1-p)}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{(1+p)}{4} & 0 & 0 & 0 \\ 0 & \frac{(1-p)}{4} & \frac{p}{2} & 0 \\ 0 & \frac{p}{2} & \frac{(1-p)}{4} & 0 \\ 0 & 0 & 0 & \frac{(1+p)}{4} \end{pmatrix}$$

$$\rho_{AB}^{T_B} \geq 0 \Rightarrow \frac{(1-p)}{4} \geq \frac{p}{2} \quad p \leq \frac{1}{3}$$

Remarks: 1) PPT state ($\rho_{AB}^{T_B} \geq 0$)

$\Leftrightarrow$ seperable state for $2 \times 2, 2 \times 3$ systems

NPPT state $\rho_{AB}^{T_B} < 0 \Rightarrow$ entangled state

2) For higher dim (such as 3×3)

exists PPT but entangled state (called bound entangled state) $E_F > 0, E_D = 0$

Hahn-Banach Theorem: If S is a compact convex set and $\rho \notin S$, here is a hyperplane to separate ρ from S .

Entanglement witness: Experimental detection of entanglement. As for separable states, PPT is also convex. So the witness is defined as a Hermitian operator W that: 1). $\forall \rho \in S, \text{tr}(\rho W) = \langle W \rangle \geq 0$. 2). $\exists \rho, \text{tr}(\rho W) < 0$. We can find that for any entangled state, there is a witness to detect it.

Ex. of Entanglement witness:

1) Entanglement fidelity to prepare (Nature 428,153(2004))

$$|\psi^-\rangle_{AB} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

set ρ_{AB} $F_{en} \equiv {}_{AB}\langle\psi^-|\rho_{AB}|\psi^-\rangle_{AB}$ $F_{en} > \frac{1}{2}$ entangled

$$W = -|\psi^-\rangle_{AB}\langle\psi^-| + \frac{1}{2}I \quad \text{tr}(\rho_{AB}W) \geq 0 \text{ for separable states}$$

2) Spin squeezing

$$\vec{J} \equiv \frac{1}{2}(\vec{\sigma}_1 + \vec{\sigma}_2) \quad J_n \equiv \vec{n} \cdot \vec{J} \quad \langle J_n \rangle = 0$$

Squeezing parameter $\xi_n \equiv \frac{\langle (\Delta J_n)^2 \rangle}{(J/2)}$ $\xi_n < 1$ state is entangled

($J = 1$ for 2 qubits)

(Nature 409, 63(2001))

3) Bell inequalities. $\langle W \rangle < 2\sqrt{2}$ (when $\rho \in S$)

Measure entanglement for mixed states: 1). $E(\rho_{AB}) = 0$ iff ρ_{AB} is separable. 2). $E(U_{AB}\rho_{AB}U_{AB}^*) = E(\rho_{AB})$. 3). $E(\rho_{AB})$ should be non-increasing under LOCC. 4). Reduce to Von-Neumann entropy for pure states.

Several typical entanglement measures:

1) Entanglement of formalism and distillation (PRL 80, 2245(1998))

E_F, E_D $E_F \geq E_D$, and $E_F > E_D = 0$ for bound entangled state

Wootters' formula for E_F of a pair of qubits

$$\rho_{AB} = \sum_i p_i |\psi_i\rangle_{AB} \langle \psi_i|$$

(ensemble decomposition)

{ p_i } Alice ————— Bob
 $|\psi_i\rangle_{AB}$

$E_C(\rho_{AB}) \equiv \min \sum_i p_i E(|\psi_i\rangle_{AB})$

entanglement cost Von-Neumann entropy measure

Over all ensemble decompositions

$E_F \leq E_C$ ($E_F = E_C$)

Def. $\tilde{\rho}_{AB} = (Y_A \otimes Y_B) \rho_{AB}^* (Y_A \otimes Y_B)$ ($Y \equiv \sigma_y$)

$$R = \sqrt{\sqrt{\rho_{AB}} \tilde{\rho}_{AB} \sqrt{\rho_{AB}}}$$

$tr(R)$: generalization of fidelity $\langle \psi | \rho | \psi \rangle$, $(tr R)^2$: fidelity between ρ_{AB} and $\tilde{\rho}_{AB}$

The eigenvalues of R in decreasing order $\lambda_1, \lambda_2, \lambda_3, \lambda_4$

Def. "Concurrence" $C(\rho_{AB}) = \max \{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}$

A measure of entanglement by itself

$$E_C(\rho_{AB}) = h\left(\frac{1 + \sqrt{1 - C^2(\rho_{AB})}}{2}\right) \text{ monotonic in } C$$

Where the function $h(x) \equiv -x \log_2 x - (1-x) \log_2 (1-x)$ (Shannon entropy)

2) Distance

$E_d(\rho) \equiv \min D(\rho, \sigma), \quad \sigma \in S$

D should be non-increasing under LOCC

Possible choice of $D(\rho, \sigma)$:

(A) $D(\rho, \sigma) \equiv \frac{1}{2} [S(\rho || \sigma) + S(\sigma || \rho)]$ [see RMP 74, 197(2002)] for properties of

$S(\rho || \sigma)$

Relative entropy $S(\rho || \sigma) \equiv tr[\rho(\log_2 \rho - \log_2 \sigma)]$

In general $S(\rho || \sigma) \neq S(\sigma || \rho)$ or ln

Property $S(\hat{\$}(\rho) || \hat{\$}(\sigma)) \leq S(\rho || \sigma)$ LOCC $\in \hat{\$}$

(B) $D(\rho, \sigma) \equiv 1 - \sqrt{F(\rho, \sigma)}$

Fidelity $F(\rho, \sigma) \equiv \left(tr \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$ Bures metric

Non-increasing under LOCC [PRL 76, 2818(1996)]

3) Negativity

$N(\rho_{AB}) \equiv \sum_i \frac{|\lambda_i| - \lambda_i}{2}$ eigenvalues of ρ_{AB}^{TB}

$\equiv |\text{sum of negative eigenvalues of } \rho_{AB}^{TB}|$

Logarithmic negativity $LN(\rho_{AB}) \equiv \log_2 [2N(\rho_{AB}) + 1]$

Upper bound $LN(\rho_{AB}) \geq E_D(\rho_{AB})$

Both N and LN are entanglement monotones. (Quant-ph 10505071)

Remark: no unique ordering of entanglement for mixed states

$$\rho_1, \rho_2 \quad \begin{cases} E_1(\rho_1) > E_2(\rho_2) \\ E_2(\rho_2) > E_1(\rho_1) \end{cases} \quad \rho_1 \xrightarrow{LOCC} \rho_2 \quad E(\rho_1) > E(\rho_2)$$

With E_1, E_2 are "good" entangled measures $\rho_1 \xleftrightarrow{LOCC} \rho_2$

such as concurrence and negativity not comparable

EPR states: $|\psi^\pm\rangle_{AB} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$.

Bell states: $|\phi^\pm\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$.

For EPR and Bell states, $E = 1$ ebit. They're locally equivalent by U_A (or U_B)

Werner state: $\rho_W = p|\phi^+\rangle\langle\phi^+| + \frac{1-p}{4}I_{AB}$. It's entangled iff $p > 1/3$.

Bell diagonal state: $\rho_B = \sum_{p, |\psi\rangle} p |\psi\rangle\langle\psi|$, $\sum p = 1$. Need to be checked by PPT.